

Final Presentation: Code Generation (ER → SQL)

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Slide 1 · Code Generation

- Goal: generate SQL DDL from bigER `.er` models
- Scope: generator part of the project
- Targets: generic SQL, PostgreSQL, MySQL
- Output: `CREATE TABLE` , primary keys, foreign keys, constraints
- Integration: VS Code commands, LSP request, CLI export path

Slide 2 · Supported ER Concepts

ER Concept	Textual	Graphical	Generator	Additional information
Entities				
Strong Entity	✓	✓	✓	Generated as a SQL table.
Weak Entity	✓	✓	✓	Generated as a SQL table with owner key columns and <code>ON DELETE CASCADE</code> when connected through a weak relationship.
Relationships				
Binary Relationship	✓	✓	✓	<code>A -> B</code> ; generated as a bridge table with primary and foreign keys.
				Supported when roles

Slide 3 · Assumption 1 — Explicit Relational Artifacts

- The generator uses deterministic ER → relational mapping rules
- Strong entities become tables
- Relationships become bridge tables
- Weak entities copy the owner key and add their `partial_key`
- Multivalued attributes become side tables
- Inheritance becomes a child table with a primary-key foreign key to the parent

Why:

- SQL output stays easy to inspect
- Golden fixtures stay readable
- Generated schemas can be validated with real database engines

Slide 4 · Assumption 2 — Constraints Where SQL Can Express Them

- Primary keys are generated from `key` attributes
- Weak entity identity is generated from owner key + `partial_key`
- Relationship participants become foreign keys
- Binary max-one cardinalities become `UNIQUE` constraints
- Weak ownership and composition use `ON DELETE CASCADE`
- Optional attributes become nullable columns

Boundary:

- Mandatory participation is not enforced in generated SQL
- We do not generate triggers or assertions
- Some ER semantics remain modeling-level information

Slide 5 · Assumption 3 — Dialects Normalize Types, Not Semantics

Example input:

```
score: DOUBLE  
photo: BLOB  
bio: CLOB  
birthday: DATETIME
```

ER datatype	PostgreSQL	MySQL
DOUBLE	DOUBLE PRECISION	DOUBLE
BLOB	BYTEA	BLOB
CLOB	TEXT	TEXT
DATETIME	TIMESTAMP	TIMESTAMP
BOOLEAN	BOOLEAN	BOOLEAN